USER INSTRUCTION GUIDE · 08 NOVEMBER 2026

FEM Automation
Composite orientation optimization

Use the supplied cantilever demo, run either stress-based method, and view the optimized FEM result in PrePoMax.

PrePoMax input → **Web app** → **CalculiX** → **PrePoMax result**

1. Find the demo files

Press **Windows + E**, paste the folder below into File Explorer's address bar, and press Enter.

Demo folder
`D:\DataBin_Composite_Automation\cantilever_beam`

PrePoMax setup model — open this in PrePoMax
`D:\DataBin_Composite_Automation\cantilever_beam\cantilever_prepo.pmx`

Exported input — upload this to the web app
`D:\DataBin_Composite_Automation\cantilever_beam\cantilever_prepo.inp`

Open the **.pmx** to inspect the beam in PrePoMax. Feed the **.inp** to the web app. The setup contains the mesh, clamp, and load; the web app adds the composite material and ply stack before solving.

2. Open the web app

In Chrome, go to http://127.0.0.1:5050/. If the page opens, continue to page 2.

If it does not open, start PowerShell and enter these two commands:

```
cd "D:\Git_Composite_Automation\fem-automation"
.\runs\.venv\Scripts\python.exe -m flask --app app run --port 5050
```

Keep that terminal open while using the app, then refresh Chrome. If the port is already in use, use the existing app instance instead of starting another one.

> **Current scope**
> Only ply orientations are optimized. Ply count, individual ply thickness, and symmetry stay fixed during a run. Use a planar shell laminate and one linear-elastic static load step.

Guide contents: page 2 — inputs; page 3 — optimization; page 4 — viewing results; page 5 — saved files, your own model, and troubleshooting.

PREPARE THE DEMO

3. Load the beam and set the inputs

1. In **Run settings**, set **Output folder** to `D:\DataBin_Composite_Automation`. Each optimization creates a new folder there.
2. Confirm that **CalculiX .exe** points to:
   `D:\PrePoMax v2.5.0\Solver\ccx_dynamic.exe`
3. In **01 Input File**, click **Load cantilever beam example**. Alternatively, upload `cantilever_prepo.inp` from the demo folder.
4. Select element set **BEAM**. For reaction-force reporting, select **FIXED_EDGE**. This node set reports reactions; the clamp itself is already in the imported input.

> **What this demo contains**
> A 100 × 10 mm shell beam, fixed at x = 0 in translation and rotation. Nine free-end nodes each carry −1 N in y: **9 N total**. Units are mm, N, and MPa.

4. Use the demonstration material

In **02 Material**, keep the following inputs for reproducing the supplied demo. The material name is **AS4_8552**.

Elastic moduli (MPa)	Poisson ratios	Shear moduli (MPa)
E1 = 135100	ν12 = 0.32	G12 = 5300
E2 = 9600	ν13 = 0.32	G13 = 5300
E3 = 9600	ν23 = 0.487	G23 = 3228.6

The five strength inputs are required for stress-based optimization:

S11t	S11c	S22t	S22c	S21s
1996 MPa	1398 MPa	64 MPa	268 MPa	92 MPa

Use validated material properties and allowables for another component. The optimization uses an in-plane maximum-stress check. The separate single-simulation section also offers an optional Puck check.

5. Set the fixed laminate in 03 Layup

Ply angles	`45/-45/90/0`
Ply thickness	0.25 mm (may display as 0,25)
Symmetric (mirror)	Checked → 8 plies, 2 mm total thickness

The full stack is `45 / -45 / 90 / 0 / 0 / 90 / -45 / 45`. The app derives the material plane from the shell mesh. A result may show −90° in place of 90°; these are equivalent fiber directions.

Sections 04 Preview .inp and 05 Simulate are optional checks of the current layup. You can proceed directly to optimization.

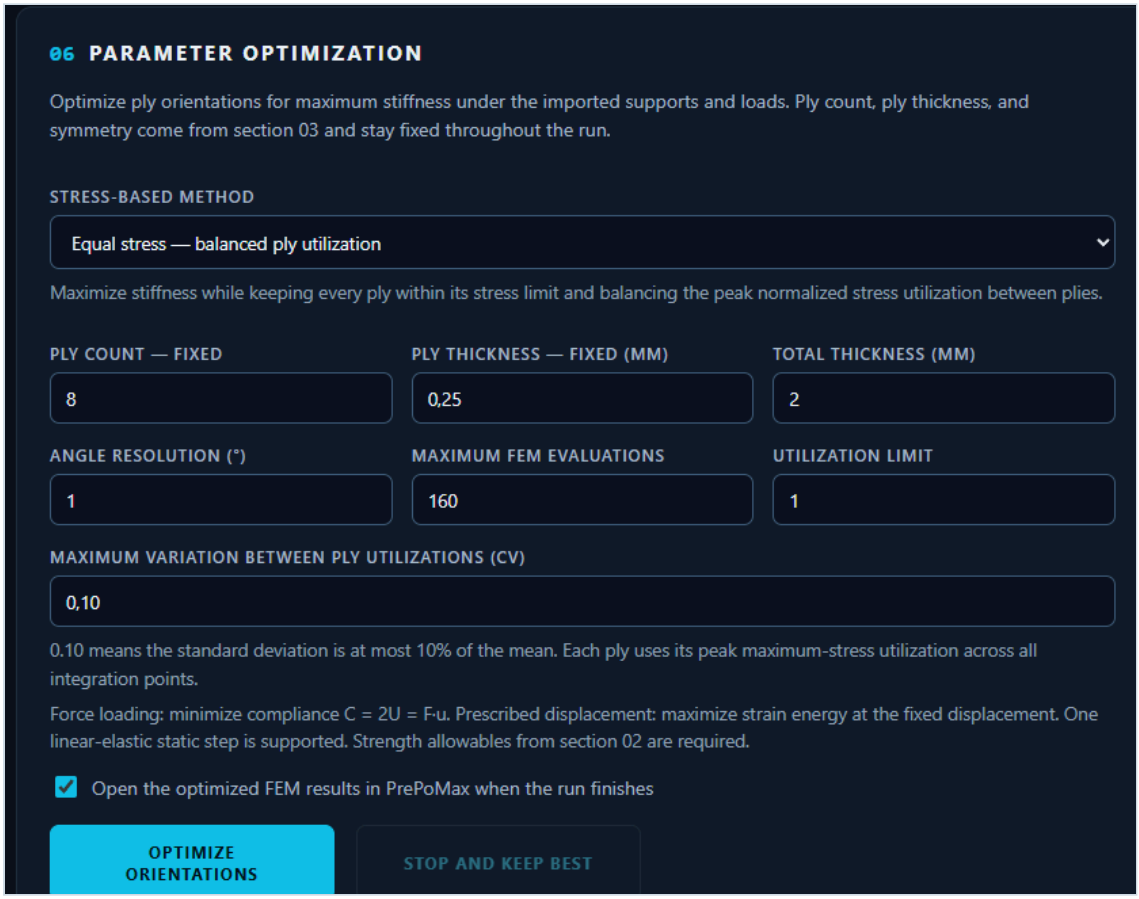
RUN THE OPTIMIZATION

6. Choose a method in section 06

Method	What it does in this stage
Fully stressed design — orientation only	Finds a stiff orientation set while keeping ply stress utilization within the selected limit.
Equal stress — balanced ply utilization	Adds a limit on variation between each ply's peak normalized stress utilization.

Use **angle resolution = 1°**, **maximum FEM evaluations = 160**, and **utilization limit = 1**. For Equal stress, use **CV tolerance = 0.10**. Check **Open the optimized FEM results in PrePoMax**.

1. Check the read-only sizing fields: **8 plies, 0.25 mm each, 2 mm total**.
2. Click **Optimize orientations**. Watch the completed-evaluation count, best-feasible chart, and evaluation history. The budget is an upper limit; a run can finish earlier.
3. Wait for **Run finished**, a feasible-design message, and confirmation that the result was verified in PrePoMax.



The method selector, fixed sizing, search settings, and run/stop buttons in the completed interface.

Compare the two methods fairly: run the second method with the original `45/-45/90/0` symmetric layup. Do not apply the optimized layup between comparison runs. **Stop and keep best** cancels the active solve and retains completed designs; use the result-opening button afterward if you want to view the retained best design.

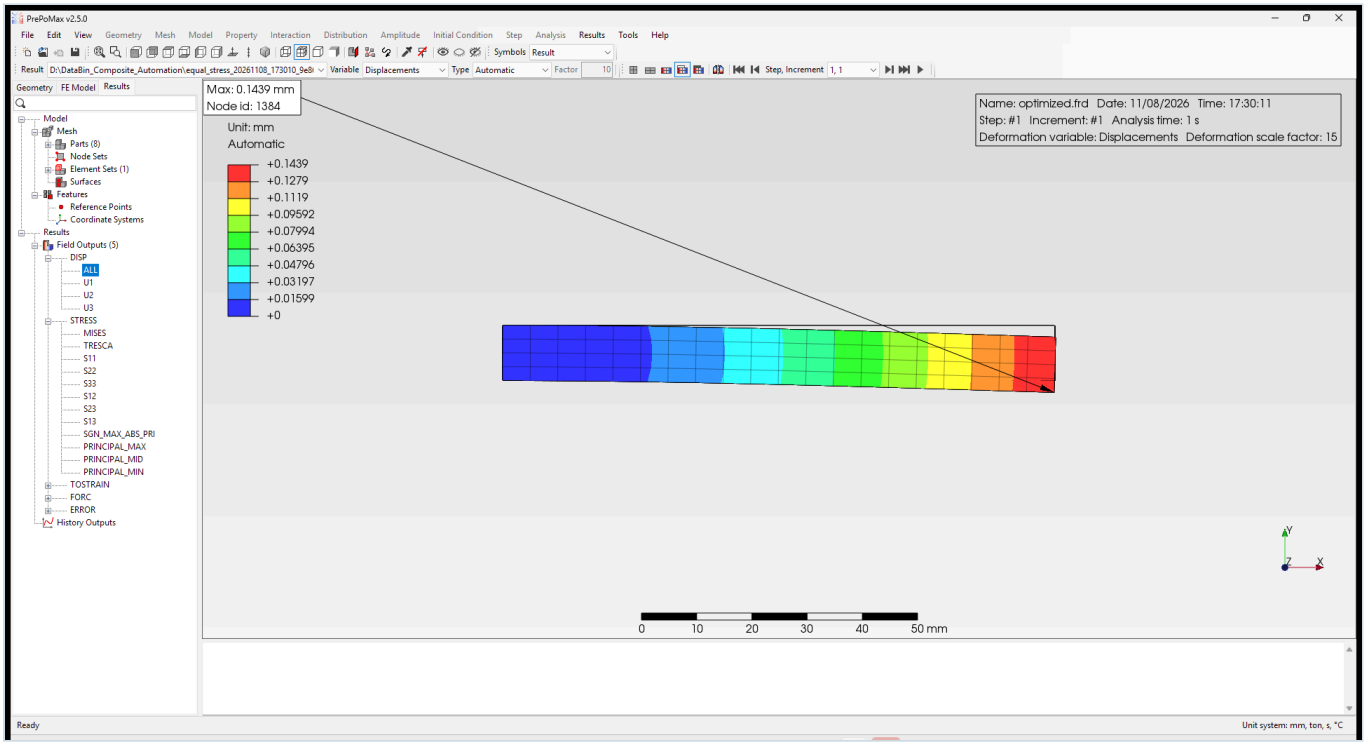
READ AND VIEW THE RESULT

7. Check the web result and PrePoMax

For this force-controlled demo, **lower compliance means greater stiffness**. A feasible-design message means the selected stress constraints were satisfied. With the supplied input, both methods found an all-0° laminate after 53 real FEM evaluations.

Quantity	Starting laminate	Optimized laminate
Compliance	3.2131 N·mm	1.29035 N·mm
Maximum reference-node displacement	0.3580 mm	0.1438 mm
Stiffness relative to the start	1×	**2.4901× (+149.01%)**
Fixed sizing	8 × 0.25 mm	8 × 0.25 mm

1. PrePoMax opens automatically when a feasible run finishes and automatic opening is enabled. Otherwise, click **Open results in PrePoMax** beneath the web results.
2. In PrePoMax, check that the **Result** selector contains the intended run folder and `optimized.frd`. Choose **Displacements** to inspect deformation.
3. Expand **Results → Field Outputs → STRESS** and select a stress component to inspect stress contours. The web ply table uses the local integration-point data for its stress check.



Verified optimized demo in PrePoMax. The displacement legend reaches approximately **0.1439 mm**. The displayed shape is magnified automatically; the deformation scale factor changes the illustration, not the result values.

Use the result: click **Use optimized layup** to copy the best angles, fixed thickness, and symmetry into section 03. Download `report.json` for the full run record or `history.csv` for the evaluated designs.

These are orientation-only adaptations. Classical fully stressed sizing would also change material sizing; this fixed-size beam remains below full utilization. A search result is local to its resolution and budget, not a guarantee of a global optimum.

FILES AND TROUBLESHOOTING

8. Reopen the demo and saved results

Use this index to locate the validated outputs and distinguish them from older diagnostic runs:

Validated results index
`D:\DataBin_Composite_Automation\CURRENT_RESULTS.json`

Fully stressed orientation run folder
`D:\DataBin_Composite_Automation\fully_stressed_20261108_172938_2bacb6`

Equal stress run folder
`D:\DataBin_Composite_Automation\equal_stress_20261108_173010_9e86a2`

Inside either validated result folder:

File	Purpose
`prepomax_viewer\optimized.frd`	Open in PrePoMax to view the optimized FEM fields.
`optimized.inp`	Complete optimized CalculiX input.
`optimized.dat`	Raw solver history and per-ply stress data.
`report.json / history.csv`	Settings, baseline comparison, constraints, and design history.
`evaluations` folder	Files from individual candidate solves.

For manual viewing, save any current PrePoMax work, choose **File → Open**, and select the FRD inside `prepomax_viewer`. That copy avoids a DAT history-reader incompatibility; the raw DAT is retained separately. Ignore folders marked **SUPERSEDED**.

9. Use your own PrePoMax object

Prepare a planar S8R/S6-compatible shell mesh with consistent normals. Define the clamp/supports and loads in a single linear static step, save the native **.pmx**, then export a CalculiX **.inp**. Upload that input, select the correct composite element set, and enter your material and fixed layup in the web app. Curved parts, nonlinear/multiple steps, and mixed force plus nonzero prescribed displacement loading need a separate formulation.

If you see…	What to do
Page cannot be reached	Start the local server using page 1 and keep its terminal open.
No feasible design found	Check material strengths, loads, and constraints; verify the CV target and evaluation budget.
PrePoMax did not open	Use the result-opening button or manually open the viewer FRD. Check the displayed viewer error.
Angle-grid or plane error	Choose a resolution dividing 180° that includes the starting angles; check the selected planar part and its normals.

Finish: wait for a run to complete or use Stop and keep best. If you started the server in PowerShell, press Ctrl+C there to stop it. Closing Chrome alone does not stop the local server.

Further instructions and method definitions are in the project's README.md and docs\optimization_methods.md under D:\Git_Composite_Automation\fem-automation.